

# Ashim Paudel

Civil Engineer

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## EDUCATION

### Tribhuvan University, Institute of Engineering, Pulchowk Campus

Bachelor in Civil Engineering

2021 – 2025

Score: **84.60%**

- Ranked **6<sup>th</sup>** nationwide among TU Civil Engineering students, 2020 batch (4-year cumulative)
- Full merit-based scholarship (4 years) — IOE Entrance Examination, **Rank 15**
- Final Year Project: Earthquake Resistant Design of Dhalkebar Hospital

### Aroma Science College, Chitwan

+2 Science, National Examination Board

2018 – 2020

CGPA: **3.93**

- Co-founded the STEM Club Aroma

## RESEARCH & PRESENTATIONS

### Poster Presentations

- “Investigating the Effects of Seismic Pounding in a Series of RC Buildings: A Case Study of the Ason Area” May 2025  
*Kathmandu Geo Lab Student Research Symposium*
- “Use of Geocells for Slope Stabilization” Jun 2024  
*Nepal Geotechnical Society (NGS) — 30th Anniversary Conference*

## EXPERIENCE

### Central Material Testing Laboratory and Consultancy Services (CMTL), Pulchowk Campus

Apr 2026 – Present

Civil Engineer

- Performed and monitored a range of laboratory tests in a standardized setup:
  - Non-Destructive Testing: UPV, Rebound Hammer, Ground Penetration Radar (GPR)
  - CBR, Permeability, Consolidation, Unconfined Compression, Direct Shear, Triaxial Tests
- Involved in Detailed Damage Assessment (DDA), design and supervision of retrofitting of:
  - Dolakha Sabha Hall, Dolakha
  - Department of Fine Arts Building, Kirtipur

### International Shake Table Competition · Asian Institute of Technology (AIT), Thailand

Jul 2025

Participant

- Represented IOE Pulchowk Campus & Nepal at “Seismic Performance of Buildings 3.0” Organized by Structural Engineering Department at AIT
- Designed a dual-system structural model of building with inner load-bearing core and outer viscoelastic damper bracing for seismic energy dissipation

### Detailed Damage Assessment and Non-Destructive Testings

Undergrad Research Assistant, RTU Pulchowk Campus

- Detailed Damage Assessment and Concrete Quality Assessment of Damaged Stilling Basin May 2025  
*Upper Tamakoshi Hydropower Project, Lamabagar Dolakha*
- Quality Assessment of Concrete Columns of Stadium Parapet Sep 2025  
*Gautam Buddha International Cricket Stadium, Bharatpur Chitwan*

**Tools:** Ultrasonic Pulse Velocity (UPV), Schmidt Rebound Hammer, SonReb, Ground Penetration Radar (GPR), Rebar Scanner

## SELECTED PROJECTS

### OpenSees Resources for Beginners

[GitHub](#) (21 Stars)

A beginner-oriented open-source repository bridging the gap between OpenSeesPy documentation and practical research application. Topics span fiber section modelling, moment-curvature analysis, and structural modelling and nonlinear seismic response of

2D/3D structural systems.

**Tools:** Python, OpenSeesPy, Matplotlib, NumPy

### Python Beam Analysis Package

[GitHub](#) (15 Stars)

A Python package for solving 2D statically determinate beams, supporting Point Loads, UDL, UVL, Point Moments, and Internal Hinges. Outputs reaction calculations, AFD, SFD, and BMD diagrams.

**Tools:** Python, SymPy, NumPy, Matplotlib

## ACHIEVEMENTS & AWARDS

|                                   |                                                                                   |      |
|-----------------------------------|-----------------------------------------------------------------------------------|------|
| <b>Rank 1</b>                     | Gandaki Province Public Service Commission Examination                            | 2026 |
| <b>Winner</b>                     | Sustainable Innovative Solutions in Civil Engineering - 3rd Aron Week Competition | 2026 |
| <b>Runner-Up</b>                  | Structural Failure Case Study Analysis Competition - 2nd Aron Week Competition    | 2025 |
| <b>Golden Jubilee Scholarship</b> | 4 years Bachelors Study Scholarship - Indian Embassy in Nepal                     | 2021 |
| <b>Mahatma Gandhi Scholarship</b> | 2 years Higher Secondary Study Scholarship - Indian Embassy in Nepal              | 2018 |
| <b>Best Club President</b>        | Leo District Council 325 E - L.Y 2023/24                                          | 2024 |

## LEADERSHIP & COMMUNITY SERVICE

**President · Kathmandu Valley Leo Club (KVLC), Pulchowk Campus**

Jul 2023 – Jun 2024

Leo Club International

- Previously served as **Vice President** (Jul 2022 – Jun 2023)

### Key Initiatives Led

- Nyano Initiatives** Dec 2023  
Led a warm clothing collection drive at Pulchowk Campus, collecting **650+ kg** of clothes from students, faculty, campus residents, and partner organizations. Donated to the Musahar community in Chepkat Village, Aurahi Municipality, Mahottari — directly benefitting **50+ families**.
- Project Paana: Library Establishment** Feb 2024  
Organized a book collection drive across Pulchowk Campus and valley schools; secured donations from publishers and organizational sponsors — totalling **1,000+ books** and stationery. Established a library at Gitawar Secondary School, Makwanpur, on the occasion of Saraswati Puja 2080.
- Prastav: Transforming Kathmandu** Dec 2022  
Coordinated a national-level ideathon addressing Air Pollution, Solid Waste Management, and Bagmati River Pollution. Engaged **25 teams** in developing actionable urban solutions, presented before policymakers including Lalitpur's Metropolitan Mayor, NAST Vice Chancellor, and other senior dignitaries.

### Other Roles

- Program Coordinator — National Shake Table Competition, Pulchowk Campus 2025  
Civil Engineering Student's Society Nepal (CESS Nepal) Exhibition 2081

## TECHNICAL SKILLS

|                                   |                                                                      |
|-----------------------------------|----------------------------------------------------------------------|
| <b>Civil Engineering Software</b> | OpenSees, ETABS, SAP2000, Bentley RCDC, Plaxis 3D, AutoCAD, Civil 3D |
| <b>Programming</b>                | Python, C, HTML/CSS, JavaScript                                      |
| <b>Data &amp; Scientific</b>      | NumPy, Sympy, Scikit-Learn, Pandas, Matplotlib, Jupyter Notebook     |
| <b>Documentation</b>              | L <sup>A</sup> T <sub>E</sub> X, Markdown, MS Office Packages        |
| <b>Version Control</b>            | Git, GitHub                                                          |

## REFERENCES

**Assoc. Prof. Dr. Kshitij C. Shrestha**

Structural Engineer

Department of Civil Engineering

IOE, Pulchowk Campus

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